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AI Agent Architecture of Artemis II's II Mission

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Agent identification

Table 1. Conceptual agents in Artemis II

The Artemis II system can be modeled as a set of interacting agents with different roles and levels of intelligence. The Flight Control Computer (FCC) operates as a model-based agent, since it maintains an internal state of the spacecraft and uses sensor data to compute control actions for stabilization and trajectory maintenance. The Guidance, Navigation, and Control (GNC) system functions as a goal-based agent, as it plans and adjusts the spacecraft trajectory to achieve mission objectives such as the lunar flyby and safe return. The Environmental Control and Life Support System (ECLSS) behave as a simple reflex agent, continuously monitoring environmental conditions and triggering automatic responses to maintain safe levels of oxygen, pressure, and temperature. The Mission Planning System acts as a utility-based agent, optimizing mission parameters such as fuel usage, timing, and safety margins to maximize overall mission success. The astronauts serve as human agents, making high-level decisions, supervising automated systems, and intervening when unexpected situations arise. Finally, ground control also functions as a human agent, providing external monitoring, analysis, and strategic decision-making support throughout the mission.

The class labels follow AIMA's distinction among simple reflex, model-based, goal-based, utility-based, and human agents [2]. The launch-vehicle row reflects Artemis II's SLS-to-Orion mission context, while the Orion and ground rows map NASA's public spacecraft and communications descriptions into agent roles [1].

Functional decomposition

Low-level reactive layer. This layer closes the fastest control loops by mapping current state error or pilot input to immediate actuation [2].

Agents: launch vehicle ascent guidance, attitude and thruster allocation control.

Mid-level state estimation and modeling layer. This layer maintains Orion's internal world model from inertial sensing, GPS, DSN tracking, star tracking, optical navigation, and telemetry [3].

Agents: Orion state estimation and navigation.

High-level planning and optimization layer. This layer chooses mission-phase transitions, burn timing, contingency actions, and safety-preserving configuration changes [2].

Agents: mission and burn manager; fault protection and safe-mode manager.

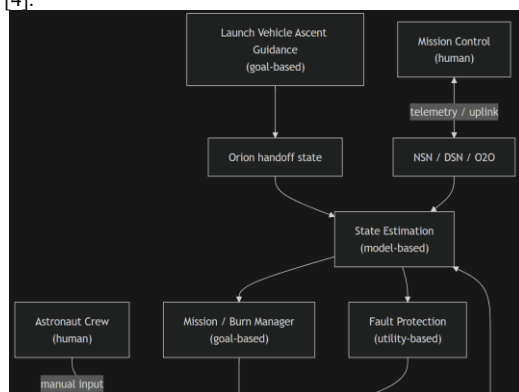
Human-in-the-loop layer. This layer adds supervisory judgment, external state interpretation, and command authority through the crew and ground networks [4].

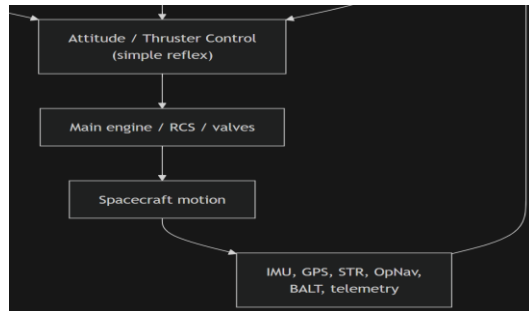
Agents: Astronaut crew; Mission Control.

Interconnections and architecture

The architecture is hierarchical because goals flow downward from mission intent to actuator selection, and distributed because sensing, control, and supervision are split across Orion, the crew, and Earth-based networks [3].

During Artemis II, ground connectivity is maintained through coordinated handoffs across the Near Space Network and the Deep Space Network [4].





These diagrams are author-generated conceptual models rather than official NASA schematics.

Perception Decision Action mapping

- **Perception:** Orion's public sensing stack includes OIMUs, GPS receivers, DSN tracking, barometric altimeters, star trackers, the OpNav camera, sun sensors, and engineering telemetry [3].
- **State estimation:** the vehicle management computers and GN&C flight software update spacecraft position, velocity, attitude, and mode from those inputs [3].
- **Decision-making:** in ALMA terms, burn sequencing is goal-based, while fault protection is utility-based because it ranks safe outcomes under degraded conditions [2].
- **Execution:** Orion's power-and-data units drive effectors such as valves, thrusters, and heaters, while GN&C commands the appropriate crew-module, service-module, and abort-system effectors [3].
- **Ground interaction:** command and data flow between Orion and Mission Control through the communication control module and Artemis II network handoffs [3].

Human-in-the-loop analysis

Astronauts act as local supervisory agents because Orion GN&C is designed to operate in both manual and automated modes and to accept pilot-control inputs [3].

They override automation when manual handling, contingency stabilization, or direct crew judgment is preferable to waiting for ground interpretation [3].

Mission Control at Johnson Space Center functions as a remote human agent because Artemis II networks carry astronaut communications, health-and-safety data, imagery, and other mission data to Earth [4].

The advantage of human participation is flexible reasoning under ambiguity, while the main limitations are onboard workload and dependence on communications for Earth-side support [2].

Failure scenario analysis

Scenario: prolonged communications loss after lunar flyby.

1. A longer-than-planned communications outage is realistic because Artemis II already expects a planned blackout when Orion passes behind the Moon [4].
2. When the Earth link drops, the state-estimation agent continues onboard propagation with inertial sensors and star trackers [3].
3. The OpNav camera supports autonomous emergency-return geometry because NASA states that it is part of Orion's lost-communications return capability [3].
4. The mission and burn manager preserves the return objective because a goal-based agent continues selecting actions that protect the desired end state [2].
5. Fault protection can shift the vehicle toward safe, power-positive attitudes by using sun-sensor cues and redundant computing paths if the outage coincides with another fault [3].

AI tools (e.g., ChatGPT) were used to support brainstorming and clarification. All final analysis and conclusions were independently verified by the author.

References

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